"Digital Defenders at the Frontline: Assessing the Issues and Dangers Faced by Organizations in their Digital and Cyber Assets"

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Many organizations still believe their firewalls and IT teams are enough to keep them safe. However, this confidence often exposes them to cyber threats (Mosteanu, 2020). Cyber threats now target home networks, personal devices, cloud services, and IoT systems. These dangers affect both business and personal environments. Basic computer and network security tools can no longer prevent such attacks (Akpan et al., 2022).

Around the world, efforts are increasing to protect IT systems from attacks that can cause severe damage. Cyberattacks on critical infrastructure are becoming increasingly common and causing significant economic losses across the public and private sectors. When key systems are hit, a nation's safety and stability may be at risk (Fedorko et al., 2017). Cyberattacks are becoming more advanced and more complex to detect (Singh & Kumar, 2020). In the past, organizations relied on firewalls, antivirus programs, and intrusion detection systems to protect their computers. These tools provided limited protection, were often used separately, and lacked coordination. This gap allowed attackers to bypass standard defenses (Muradyan, 2023).

According to Li and Liu (2021), a strong dependence on technology increases the exposure of digital assets to various threats. Cyber threats encompass a range of

malicious activities in cyberspace, including website defacement, espionage, theft of intellectual property, denial-of-service attacks, and destructive malware (Abu et al., 2018). Cyber threats began during the early years of computing. Early attacks included simple viruses and worms that affected only personal computers. By the 1980s and 1990s, as Internet use spread, attackers started targeting networks and larger infrastructure systems (Akeiber, 2025).

In recent years, more incidents of digital asset breaches have led to significant data and resource losses. These violations pose serious risks to individuals, organizations, governments, and society (Wang et al., 2024). Both individuals and groups are affected by cyberattacks. Stolen identities, shame, and blackmail are what one person faces because of these attacks. The other person faces trust and legal issues, as well as the business's closure, due to the cyberattack. System crashes and stolen information erode public trust and could cause the country to fall out of whack. These rules show how urgent it is to improve the type of products and ensure that they are selected (Akpan et al., 2022).

Schools are facing over 3,000 cyberattacks each week. One standard method used by malicious individuals is phishing. Every day, 34 billion phishing emails are sent worldwide. According to IBM (2024), the world's average cost of a single data breach has now exceeded $4.88 million. Almost one in four people has been affected by password-based attacks in recent years. Stolen login details for banking and financial accounts are sold for an average of $70.91. A 7-character password, even with numbers, uppercase letters, lowercase letters, and symbols, can be cracked in just 4 seconds (Kumar, 2025).

Cybercriminals are increasingly targeting less developed and developing countries, including the Philippines, which ranks among the nations with the highest number of ransomware incidents (Al-Rimy et al., 2018). Because it depends heavily on technology, the Philippines is more open to cyber risks (International Trade Administration, 2020). The Philippine National Police (PNP) reports stopping about 1,200 cyberattacks each day. Studies

also show that the country has one of the highest global rates of online child exploitation. It also faces significant challenges in protecting its citizens' biometric data (Untalan, 2025).

These findings highlight the urgent need for stronger cybersecurity systems, better data protection, and greater public awareness nationwide. There are about seven pending House bills in the Philippines on cybercrime. However, aside from the E-Commerce Law, no other law focused on Internet activity has been passed. Many issues make the problem worse. These include a limited budget for cybercrime prevention, a lack of a central coordinating agency, poor coordination between government and private groups, untrained law enforcers, and low public awareness. Effendy Ibrahim, an Internet safety advocate and head of Symantec Asia's Consumer Business Division, called the situation a "silent epidemic." He added that many Filipinos remain unconcerned about the growing cybercrime cases (Li, 2021).

Cybercrime continues to pose a significant problem for both the public and private sectors in the Philippines. This study examines the issues and threats that organizations face in protecting their digital assets in Nasipit, Agusan del Norte. It also reviews the strategies used to defend against attacks and assesses the effectiveness of current security policies. The study also looks at how people, training, and planning can improve digital protection. This research is vital because organizations now rely heavily on digital tools and data while facing constant danger from hackers and cyberattacks. Cyber-attacks can steal your information, cost you money, and damage an organization's reputation. Cybercriminals are constantly finding new ways to use the same threats, such as ransomware, phishing, and malware, in more complex ways. The research helps in clarifying the various cyber risks. It also provides guidance on strengthening defenses and safeguarding key information. In general terms, this study emphasizes the need to help organizations develop safe and resilient systems.

1.2 Review of Related Literature

The reviewed literature focuses on identifying problems, risks, and strategies used by digital defenders to protect organizational assets and maintain security amid growing cyber challenges. These threats continue to evolve, making cybersecurity a significant concern for institutions and organizations.

WEAK CYBER GUARDIANSHIP

The growing trust in telecommunications and the Internet has led to an increase in technology-facilitated crime, or cybercrime. As formal agents of social control, such as the police, have struggled to address criminal offenses in cyberspace, informal guardianship has been activated in practice in the field of study. In line with this, Vakhitova, Go, and Alston-Knox (2023) argue that informal guardianship has long been understood to refer to established crime and remains an essential process for prevention and involvement in cyber abuse incidents. Using mixed methods that combined thematic analysis and logistic regression with Bayesian variable selection, the researchers analyzed data from a sample of US adults who reported experiences of cyber abuse.

According to Vakhitova, Go, and Alston-Knox (2023), informal guardianship, once associated with physical crime prevention, also plays a key role in stopping online abuse. Their study used both qualitative and quantitative methods, combining thematic analysis with logistic regression and Bayesian variable selection. The researchers examined data from US adults who had experienced cyber abuse. The findings showed that family members, friends, partners, coworkers, online peers, and even regular Internet users often serve as informal guardians in the digital space. They also discovered that a person's sense of responsibility as a guardian significantly affects their chance of intervening. Frequent social

interaction among these groups increases both the willingness and opportunity to act against online abuse.

POOR CYBERSECURITY POLICIES

The Philippines continues to struggle with major cybersecurity issues caused by outdated policies and weak enforcement. According to Li (2021), the country's high rate of cybercrime indicates weaknesses in its national cybersecurity system. Frequent cyberattacks disrupt both government and private institutions, proving the state's limited ability to maintain strong defenses. While the use of e-services has expanded to improve public service, the lack of consistent privacy and data protection measures still puts national security at risk. Li also noted that untrained law enforcers and poor coordination among agencies make it hard to identify and prosecute cybercriminals effectively.

Brucal, Abante, and Vigonte (2025) stated that the Cybercrime Prevention Act of 2012 (R.A. 10175) was outdated and did not work effectively in today's day and age. There are fuzzy words and weak ways of keeping people from doing bad things like stealing someone's identity, committing internet fraud, and hurting children online. What is more, the law has been used in ways that create conflicts between keeping computers safe and people's rights. A good example is the case of journalist Maria Ressa, who was found guilty of cyber libel. These problems show that it is still hard to keep a good balance between keeping the country safe and keeping the Internet free.

The authors further pointed out that poor inter-agency coordination and limited cybersecurity training continue to weaken the country's digital resilience. Both reports confirm that the Philippines' cybersecurity weaknesses stem from outdated laws, weak implementation, and limited institutional capacity. Uncoordinated cybersecurity measures, combined with low public awareness, make it harder to prevent cyberattacks. As technology rapidly advances, the Philippines must strengthen its cybersecurity policies, improve inter-

agency coordination, and invest in capacity-building to create a safe, secure digital environment.

According to the DICT, only 55 of the 388 government agencies responded to notices concerning weaknesses in their systems or applications. Thus, compliance with policies on system weaknesses is very low (NEDA, 2024). This highlights systemic weaknesses in policy enforcement. Only 6% of organizations in the Philippines are fully prepared to address cyber threats (Cisco, 2025). Many still lack updated systems and trained cybersecurity personnel.

Furthermore, about 35% of companies still use outdated defense tools, which makes them vulnerable to hacking (NTUC Philippines, 2021). Budget cuts of over ₱300 million compound the problem by limiting the government's ability to spend on cybersecurity. Insufficient numbers of skilled personnel impede the proper implementation of policies (Philstar, 2023; Manila Bulletin, 2024).

WEAK PASSWORD POLICIES

In terms of cybersecurity negligence, there are many types; some are obvious, and some are not, but all pose risks that companies should not take. Here are the most common and deadly negligent actions that make organizations open to the most serious security issues, and why these can lead to costly failures.

Passwords are usually the one thing that gets you into any system. Often, organizations take on their own risk by allowing users to use weak defaults and/or easy-to-guess passwords, and by failing to implement reasonable security practices such as username verification. A weak password policy would allow users to pick a password like "admin" or "password123" and never change it. Also, many organizations do not deactivate orphaned accounts that are not currently in use, creating new access points that could be used.

VIRUSES

Cybersecurity threats are evolving as new technologies emerge and bad actors learn more online. People, organizations, and even nations face these threats as they become more complex and widespread. In computing's early years, cyber threats were basic and not very widespread. One of the first examples of the Creeper Virus and the Morris Worm was that they could hop from machine to machine. Even though the programs were experimental, they paved the way for bad things. As computer networks expanded in the 1980s and 1990s, attackers began to exploit the growing connectivity between systems. This period saw the rise of early malware that caused damage and compromise to computer systems (Lutkevich, 2022).

This thesis examines penetration and ways to enhance security in the defense sector. As cyberattacks become more sophisticated, securing national security has become more critical. The research identifies the weaknesses that make the defense sector more vulnerable to cyber-attacks. The data were collected through semi-structured interviews with cybersecurity experts, which revealed how the threat landscape is changing. The findings highlight several key themes: the rising sophistication of cyberattacks, the importance of proactive threat detection, and the need for stronger cooperation. An analysis of previous cases and current trends shows that the security of the defense industry has achieved some triumphs but also faced challenges. The research concludes with strategic suggestions to strengthen the industry's protections, such as using more sophisticated tools, encouraging round-the-clock learning, and improving rulemaking (Lee, 2024).

Today, we live in a world where technology is so ubiquitous that every business needs to protect its knowledge and information assets. Cybersecurity in Knowledge Management: Cyberthreats and Solutions tells us how knowledge management and cybersecurity are linked. It provides a simple overview of the steps, tools, and systems required to protect organizations' knowledge. As cyberattacks grow more advanced,

especially in fields such as digital marketing, supply chains, and higher education, the book discusses how organizations can strengthen cybersecurity without limiting creativity or flexibility. It introduces an integrated framework for securing knowledge in modern networked environments, using insights from artificial intelligence, machine learning, and human behavior. It also reviews the use of AI and machine learning in cybersecurity and compares different methods for classifying malware. In addition, it includes case studies on data breaches and insider threats within knowledge systems. Finally, it examines emerging cybersecurity challenges in Africa and explores how culture and social engineering influence knowledge protection within organizations.

The book also highlights key topics, including the cybersecurity needs of online marketing, the effects of COVID-19 on knowledge sharing in higher education institutions, and the importance of regulatory compliance and cross-industry collaboration. Cybersecurity in Knowledge Management: Cyberthreats and Solutions is designed for experts, researchers, and policymakers. Its interdisciplinary approach makes it valuable for understanding how cybersecurity and knowledge management intersect across different sectors.

The rise of digital technology has improved many aspects of life but has also made cyber threats more complex. Personal data has become a primary target for attackers due to its high value and potential for misuse. Emerging threats now include phishing, malware, ransomware, data leaks, attacks on Internet of Things (IoT) devices, and manipulation through artificial intelligence and deepfake technology. These dangers threaten individuals, organizations, and the overall trust in digital systems. As cyber threats continue to grow, protecting personal data remains a significant concern. Effective strategies include using two-factor authentication, encrypting data, keeping systems up to date, and applying artificial intelligence tools to detect and prevent attacks early.

Furthermore, it is vital to enhance digital literacy among the public, as it is a prerequisite for combating social manipulation and addressing legal issues related to data privacy, such as those within the scope of the Personal Data Protection Act. A better understanding makes people more comfortable with their right to privacy. A protected environment in the virtual world can be achieved through a combination of technical measures, appropriate legislation, and improved public education about digital behavior. As Widjaja (2025) states, research into interventions examines the existing online threats and explores how they can be mitigated using current technological services.

RANSOMWARE ATTACKS

There was a significant turning point in cybersecurity in the 2000s. During this decade, cybercrime exploded. This was mainly for the sake of money. The hacking community made malware more sophisticated by developing keyloggers and Trojan horses for banking purposes. Cybercriminals built botnets to perform large-scale work, such as distributed denial-of-service (DDoS) attacks.

Ransomware has become one of the most serious and fast-growing threats to cybersecurity. It became a significant problem in the middle of the 2010s, when the number of attacks began to grow rapidly. In a ransomware attack, the attacker encrypts the victim's files or systems and demands money to restore them. These cases have hit both local and foreign targets, showing how common and dangerous ransomware can be (Lee, 2024).

In May 2017, people watched WannaCry ransomware spread rapidly worldwide, infecting over 500,000 computers across 150 countries (National Cybersecurity and Communications Integration Center, 2023). WannaCry exploited one of the most common Windows bugs to spread. This bug was called EternalBlue. The US National Security Agency (NSA) built it. A group of hackers called the Shadow Brokers obtained it after breaking into the NSA's computers and released it. When the malware was on a computer, it would lock

files, set them to read-only, and demand a Bitcoin ransom to unlock them (Yadav et al., 2019; Williams, 2020). The global spread of WannaCry was huge, affecting vital services, health care, and firms worldwide. Its broad reach showed how destructive ransomware can be when it targets critical services. It also received extensive media coverage due to its scale and the damage it caused (Lee, 2024).

Ransomware attacks remain, by far, the most damaging cyber threat we face today. A good example of just how damaging a hacker attack on a key system can be is the Colonial Pipeline attack in May 2021. Colonial Pipeline is one of the largest fuel lines in the US. It carries much of the Gulf's fuel to the East Coast. In this case, the hackers used ransomware called DarkSide to stop the company from operating. This caused fuel shortages and price hikes in several East Coast states (US Department of Energy, 2021). Attackers used flaws in the Colonial Pipeline's system. They locked up the company's data and demanded a ransom to unlock it. To address the fuel supply and delivery problem, Colonial Pipeline temporarily halted operations. The event revealed the real links between critical infrastructure systems and how a single cyberattack could disrupt essential public services (CISA, 2023).

Ransomware attacks are grave and increasing danger. They cause significant disruptions and financial losses for both companies and individuals. The Verizon 2023 Data Breach Investigations Report (DBIR) showed that ransomware made up 24% of all breaches (Hylender et al., 2024). Sophos (2023) further said that 66% of companies faced ransomware attacks. The rise in ransomware, along with more sophisticated cyberattacks, poses a significant threat to overall cybersecurity.

There has been a rise in cyberattacks in the Philippines, mainly ransomware attacks, which accounted for 62% of all attacks worldwide in 2020. The same type of hack hit at least 7,000 businesses nationwide, according to a 2021 Manila Times story. A company from the Philippines that just had this bad luck is S&R Membership Shopping, which was infiltrated and had about 22,000 people's details stolen (CNN Philippines, 2021). A second story noted

that groups working in the Philippines perform worse than the global average at preventing hackers from locking data in place (Rappler, 2021). The Philippine Health Insurance Corporation recently had the same kind of hack and had to pay a group of famous hackers $300,000. PhilHealth learned of the hack on September 22, 2023, and subsequent checks indicated that the Medusa group likely carried it out (Samaniego, 2023).

Ransomware attacks have risen significantly worldwide and pose a significant risk to digital systems. The attacks hit a variety of victims, from persons, critical infrastructure like power plants and health facilities, to major corporations and government bodies. Attackers use sophisticated encryption methods to steal data and demand large ransoms for the keys to unlock it. These stringent procedures make it challenging to recover data without the attacker's secret key, with serious consequences for victims. Early identification of dangerous behavior, ideally before any damage occurs, is key to reducing these risks. Although many research works have focused on ransomware and offered detection and prevention solutions, the body of literature still lacks effective early-detection methods. It addresses the challenges and constraints of existing methods (Mingcan Cen et al., 2024).

CYBER WARFARE

In the 2010s, cyber threats grew and spread worldwide. For example, Stuxnet, found in 2010, shows how countries use cyber war (Lee, 2024). This worm is tricky. It attacked systems that control tools and data, especially those that Iran used at nuclear plants. It infected machines that handle uranium, causing considerable damage and slowing Iran's work with nuclear stuff. Stuxnet was the first cyber weapon to hit any key system so hard. The strike showed that cybercrime can be wielded as a powerful tool by the world, sparking debate over the use of cyber weapons in global conflicts (Raghunandan, 2022). Besides this, the 2016 attack on the Democratic National Committee (DNC) put cyber risks at the center of world talks.

In the US, the election for president had an issue where Russia took and gave out files about politics. The disclosure of Russia's involvement sparked discussions about other countries' interference in elections. It also raised questions about how hacking changes discussions and votes around the world. This event showed how hard it is to identify the exact country responsible for the hacking and what it may mean for foreign relations (McMaster, 2020).

The danger from online crime has risen slowly, and it is now higher and more common. The FBI's Internet Crime Complaint Center (IC3) has records of 3.26 million complaints from 2018 to 2022, costing 27.6 billion dollars (Lee, 2024). The rise was driven by online criminals becoming more proficient and adopting new tactics, such as malware, ransomware, and phishing, to breach weak systems and networks.

Cyber war is a new and big idea now. They say it is the fifth war side after land, sea, air, and space. Cyber war can hurt. They can turn off soldiers' email, blow up mines or lines, stop planes from talking to each other, make trains go wrong, send wrong bank data, cut power, and break satellites.

Cyber warfare has one of the worst problems: it is hard to find out who is attacking and who they are. Cyberwar can also have the same results as nuclear war. It also shows how important it is to have security systems and effective procedures to protect our communication systems and computers. Because of these threats, cyber defense is a big part of homeland security (Fatmir Basholli, 2022).

AI-POWERED ATTACKS

The rise of AI in hacking is a real danger for companies. Bad guys are now using AI to make attacks that are really clever and hard to see. Many tools, such as FraudGPT and WormGPT, allow hackers to create highly detailed, continually evolving attacks (BDO Digital,

2023). These attacks can target humans and computers and use AI to find ways in. AI that operates autonomously can learn to get in, hide from those trying to stop it, and stay hidden, making it very hard to fight back (Pearson, 2024).

The easy to use tools that use ai to hack dark forums, dark markets and the dark web has made it easier for for people to sell the more challenging to catch bad things on the web even people with no skills can use the crazy ai features it has made the bad people bigger beasts in the web and by doing this the people that use it for bad has no tech skills all they do is buy the tools from the dark web that are hacked kits and malware they are to lazy to make. They buy it and run it on your website. They do this because it makes it easier for lazy people to do it. They have a easy way to hack on your web by buy from the dark web instead of having to make the tool from scratch this is way worse for the good people because it makes them lazy and dumb for the work they need to do so than those people make the money and make this world a worse place besides the good people with skills and skill and having to make it from scratch.

AI-Lex attacks differ from traditional cyberattacks because they can bypass tools that rely on rules and scripts. These help to find the threat. This makes it harder to detect and block (Yamin et al., 2022). A significant risk of these AI-led attacks is that they can learn and adapt to bypass new safety measures (BDO Digital, 2023). Unlike older safety tools that use a single set of rules to detect threats, these tools can adapt as safety improves. This means firms need to regularly check their safety and update it often to stay protected from new risks (Yamin et al., 2022).

According to Kaspersky Security Network (KSN) findings, the Philippines moved from 76th place on the list of countries most affected by local cyber threats to 66th. In response to the increased cyber threats in the Philippines, there is a need for stronger cybersecurity measures. In 2024, the Philippines had 17,706,118 local cyber threats affecting 36.8 of its users, or approximately 48,500 Cyber Threat cases per day. The total number of

Cyber Threat cases decreased from 22,731,157 in 2023; however, the Philippines' overall cybersecurity level declined internationally (Amojelar, 2025).

The growth of AI use in cybersecurity brings both risks and benefits to the field. AI could be used to do many things that are not good, but it could also be used to improve how we do things in safer ways. The bad guys have used it to do massive, fast, and complex attacks. This paper shows how AI can put you at risk, including fake photos, AI phishing, automated attacks on weaknesses, and robots creating malware. It also explains new attack methods, such as prompt injection, model stealing, and data poisoning, which pose a threat to AI safety. Drawing on the latest research and real-world studies, this paper explains how these threats occur and what can be done to stop them. These ways include making AI more transparent (XAI), identifying what is wrong (anomaly detection), and using AI to train people to be aware of risks. This paper aims to help the security group learn about and avoid these AI-related dangers, and to show where to work together to stay ahead.

The growth of AI has changed how we approach cybersecurity and how bad guys find holes in our systems. The guys who do bad things keep looking for things we do not yet know about. We call these zero-day attacks. A zero day is not known to us. It is hard to watch out for these zero-day attacks. They happen before those who watch our back can stop them. Old systems that detect break-ins mostly look for known attack signs, so they cannot detect or stop zero-day threats.

Machine learning (ML) methods can be practical tools for strengthening IDSs by detecting malicious activity in real time and predicting potential weakness spots. This paper examines how the most advanced AI technologies, such as ML, DL, and FL, can combat a class of attacks known as zero-day attacks. It explores how AI can help guard against hacks that exploit weaknesses we do not yet know about.

The document expands on the training data for AI systems and on experiments with different algorithms on actual cyber threats. The key issues in using machine learning for zero-day detection include balancing the training data, developing a universal method for identifying multiple threat types, and creating cost-effective yet reliable solutions. The report concludes with recommendations to enhance AI protection and adopt proactive measures to address future cyber threats.

TRAINING AND AWARENESS

Taherdoost (2024) said hackers change frequently, making it harder for groups and individuals. This shows we need to learn to stand up to bad people. To do this, the paper developed the Integrated Cybersecurity Awareness Training (iCAT), a new approach to learning cybersecurity.

The iCAT system combines knowledge graphs, serious games, and gamification elements to make learning both fun and effective. Its micro-learning units provide brief, adaptable lessons that fit into any schedule. Additionally, the system includes instant progress tracking and personalized feedback, allowing real-time interaction and customization to each learner's needs during training.

Test results for the iCAT system indicate a marked increase in engagement and understanding compared to standard security awareness measures. Its adaptable, user-friendly design makes it a practical, manageable tool for organizations seeking to enhance cybersecurity through user-focused strategies.

TECHNOLOGICAL DEFENSE SYSTEMS

From 2020 to 2024, China's online rules in the Indo-Pacific shifted from cautious to at the forefront. This move is part of a new approach to stopping hackers, known as cyber

deterrence. It has three parts to make hackers think twice. One part is to use sharp tools that stop people from trying to break in. The other two parts are the rules China has for its own safety. This new system not only keeps China's online world safe but also helps China have a voice on the global stage through the Net (Li, 2024).

China's new tech shield is all about AI, quantum, and 5G. These tools can add to and keep their cyber stuff. They can now track, keep, and guard their servers and network more safely. China's use of AI to stop hackers can now better and faster warn, predict, and block bad things. Quantum can let linkies that cannot get hacked move info fast enough to beat the hackers. Also, finally, 5G can give China the fast speeds and tight grip on info that make it the next big thing in digital around the world (Li, 2024).

These tools support China's stringent rules, such as the Cybersecurity Law (2017) and the Network Data Security Regulation (2024). These rules give China a formidable defense, closing off access and imposing heavy limits on the flow of information. They are key to storing data at home, restricting internet access, and keeping the state firmly in control. By keeping out foreign access, especially from America and its friends, China boosts its digital shield. The study also notes that this approach is a stance of deterrence by denial, blocking anyone who might want to breach China's cyber defenses (Li, 2024).

The Cyber Defensive Policy of China is focused on working with other countries internationally to build relationships, as evidenced by China's advocacy for a policy of cyber sovereignty and its initiative, The Digital Silk Road. This has been done in part to create a dependency on China's ability to provide its digital standard, therefore creating a technological dependency that supports and enhances China's cyber power. Additionally, China has utilized its diplomatic and legal means, in conjunction with technology, to build a cyber-technology barrier that exists within the region; this area (in this case) is the Asia-Pacific Region and limits the options available to nations in that area to counter the efforts

of the United States and its allies in the Indo-Pacific region, particularly through initiatives such as AUKUS, Quad, and the Secure Equipment Act of 2021 (Li, 2024).

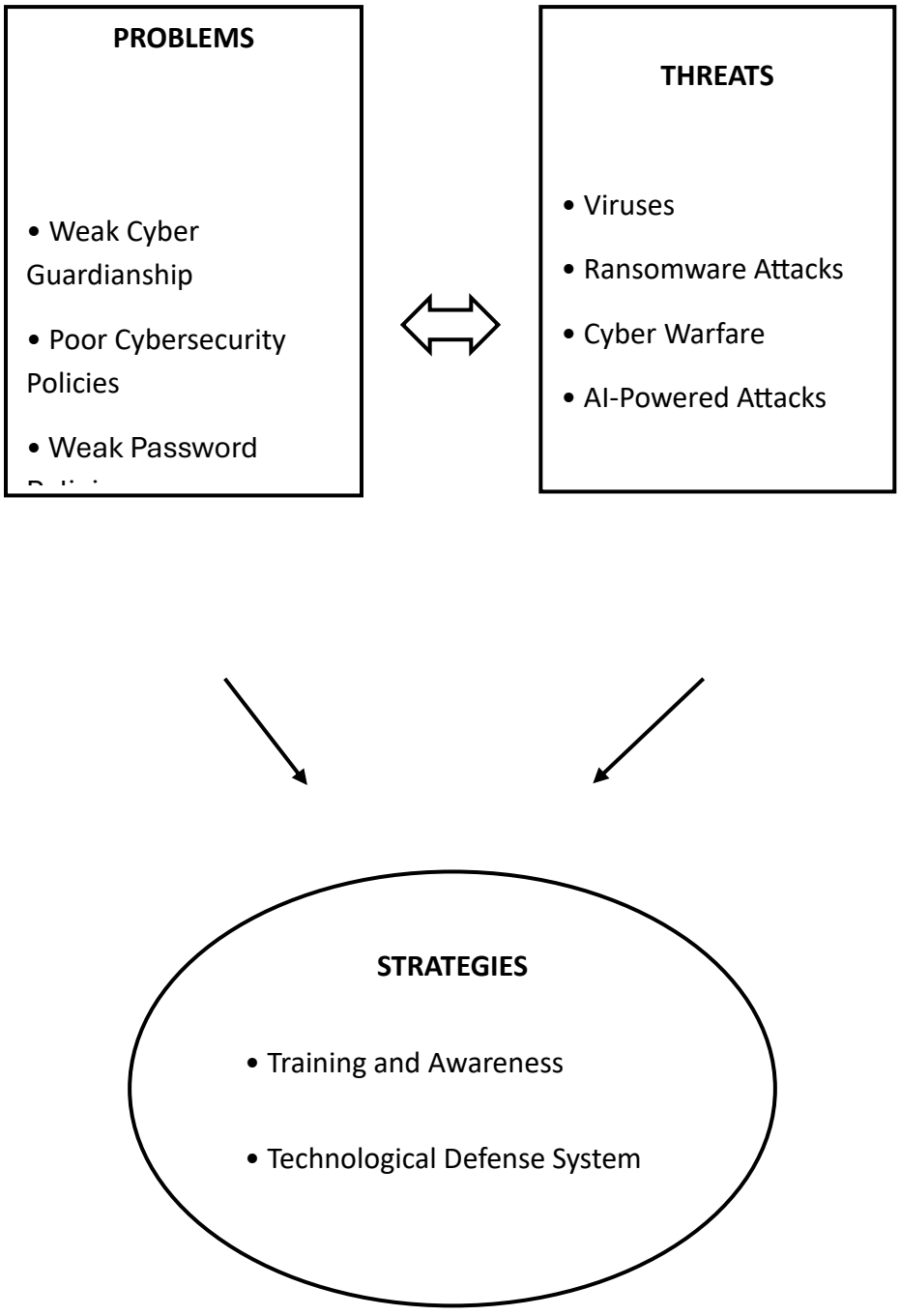
Theoretical Framework

This study is anchored in Routine Activity Theory (RAT) by Cohen and Felson (1979), which explains that crime occurs when three conditions converge: a motivated offender, a suitable target, and the absence of a capable guardian. In the digital world, RAT helps explain how cybercrimes arise from everyday use of technology. For organizations and small businesses, routine online activities such as digital transactions, customer messaging, managing Facebook pages, and using online tools create many opportunities for cybercriminals seeking to exploit weaknesses for profit or unauthorized access. Digital systems, business pages, customer records, and online accounts become "suitable targets" when they have vulnerabilities such as weak passwords, outdated security settings, or insufficient monitoring. The "capable guardians" in cyberspace are the people and tools that protect these online assets, including IT personnel, cybersecurity officers, business owners themselves, and security measures such as two-factor authentication, firewalls, and proper account management. When these guardians are weak, untrained, overworked, or unsupported, cyber threats become more likely.

Applying RAT to the study "Digital Defenders at the Frontline: Evaluating the Problems and Threats Experienced by Organizations in their Digital and Cyber Assets" helps explain why cyber threats happen and why many individuals and businesses remain vulnerable. The study includes respondents who run small businesses in Agusan del Norte and use online platforms, especially Facebook pages, to sell their products, making them common targets of scams, hacking, and other forms of online fraud. The first objective, which identifies the cyber threats and problems these organizations and online sellers face, is tied to how offenders exploit weak security. The second objective, which examines the factors that make them vulnerable, focuses on daily online activities, a lack of cybersecurity

knowledge, and limited digital protection. The third objective, which analyzes the experiences of digital defenders and online business owners, highlights how they act as guardians and the challenges they face in protecting their digital assets. Overall, Routine Activity Theory helps the study understand how everyday online activities, weak digital protection, and the experiences of both cybersecurity personnel and online business owners in Agusan del Norte affect the safety of their cyber assets.

1.3 Schematic Diagram



1.4 Objectives of the Study

1. To identify the various types of cyber threats and security problems encountered by organizations in managing their digital and cyber assets.

2. To examine the underlying causes and contributing factors that make organizations vulnerable to digital and cyber threats.

3. To analyze the lived experiences and perceptions of digital defenders (e.g., IT personnel, cybersecurity officers, and data managers) regarding their roles and challenges in protecting organizational assets.

1.6 Scope and Limitation

This study focuses on the cyber threats and security problems experienced by small business owners in Agusan del Norte specifically in Buenavista, Nasipit and Carmen who use Facebook pages to sell their products. It looks at the types of online threats they face, the reasons why their pages become vulnerable, and their personal experiences in protecting their digital business. The study includes only eight respondents who were chosen because they actively manage their Facebook business pages. Through interviews, the study aims to understand their challenges and how they deal with different online risks.

However, the study also has limitations. It includes only eight people, so the results cannot represent all online sellers in the area. The study is also limited to businesses within

Agusan del Norte and may not apply to other places. It focuses only on Facebook sellers and does not cover other platforms like Shopee, Lazada, or TikTok Shop. The information depends on what respondents remember and are willing to share, which may not always be complete. Lastly, since the study uses a qualitative approach, it provides detailed stories and insights but cannot give broad statistical results.

1.7 Definition of Terms

AI (Artificial Intelligence) Powered Attacks. Cyber threats involving AI systems, which are seek out and then exploit weaknesses.

Cybercrime. A malicious act that is conducted through any digital medium, including but not limited to hacking, identity theft, and online fraud.

Cybersecurity. This concept describes the field that focuses on the safeguarding of systems, networks, and data that reside within cyberspace against various attacks, unauthorized access, or data destruction.

Cyber Guardianship. People and technology (e.g., IT staff, firewall, and training programs) are engaged to defend an organization.

Cyber Threats. Any malicious effort to destroy, steal, or disrupt the data and systems, which may involve malware, ransomware, or phishing.

Cyber Warfare. The conflict in which a nation or organization engages in cyberattacks to destroy, disrupt, or degrade another entity's capacity.

Digital Assets. Any significant digital asset that an organization owns. Digital assets consist of data, software, databases, online platforms, or similar resources.

Digital Defenders. A group of individuals or teams responsible for safeguarding against cyber and digital threats to an organization.

Malware. Any software purposely designed to interrupt, destroy, or gain unauthorized access to a single computer system.

Ransomware. A type of malware that disables or encrypts data and requires payment for the restoration of data access.

Vulnerability. Systems, processes, and individuals that a cyber attacker can exploit.

CHAPTER 2

RESEARCH METHODOLOGY

2.1 Research Design

This study used a qualitative research design to explore the cyber threats and security problems experienced by small business owners who use Facebook pages to sell their products. Qualitative research is appropriate because it focuses on understanding people's experiences, opinions, and personal stories rather than numbers or statistics. Through this design, the researchers were able to gather detailed explanations about the types of cyber threats encountered by the respondents, the reasons why they became vulnerable online, and their experiences as "digital defenders" of their businesses. The study used semi-structured interviews as the main method of data collection. This type of interview allows the researchers to ask prepared questions while still giving respondents the freedom to explain their answers in their own words. This helped the researchers explore deeper insights, emotions, and real-life situations faced by the participants. Because cyber threats and digital security issues are personal and often sensitive, a qualitative design allowed the respondents to speak openly and comfortably about their experiences.

2.2 Research Materials and Methods

This study uses a qualitative research design, specifically semi-structured interviews, to understand the cyber threats and problems experienced by small business owners who use Facebook pages for selling products. A qualitative approach is used because it allows the researchers to gather detailed stories, personal experiences, and insights from the respondents. This method is appropriate for exploring how online sellers experience cyber threats, what makes them vulnerable, and how they defend their digital assets. It also helps the researchers understand the emotions, challenges, and real life situations faced by the respondents. Additionally, existing documents or materials related to cybersecurity practices could be analyzed to provide further context. Participants will be selected through purposive sampling to ensure they have relevant knowledge or experience. The collected data will be carefully coded and analyzed to identify common themes and patterns. Ethical considerations, such as informed consent and confidentiality, will be strictly followed throughout the research process, ensuring the respectful treatment of all participants and the integrity of the study. This combination of methods will allow the research to uncover detailed insights that numerical data alone cannot reveal.

2.3 Research Environment, Population and Sample

This study was conducted in Agusan del Norte, a province where many small business owners use Facebook pages to promote and sell their products. The area includes both urban and rural communities, and online selling has become a common way for residents to reach

customers. In this environment, business owners depend on digital tools such as Facebook, Messenger, mobile banking, and online advertisements. Because they rely heavily on online platforms, they also face online risks such as scams, hacking attempts, and fake transactions. This makes Agusan del Norte an appropriate place to explore the cyber threats and challenges experienced by digital defenders specifically small entrepreneurs managing their own online business pages.

The population of the study includes small business owners in Agusan del Norte who use Facebook pages as their main platform for selling products. These individuals handle their own digital operations, such as posting products, managing messages, handling orders, and securing customer information. They are considered “digital defenders” because they protect their online business pages from cyber threats and suspicious activities.

The sample of the study is composed of eight (small online business owners selected from different areas within Agusan del Norte. They were chosen through purposive sampling, meaning they were selected because they have direct experience in managing Facebook business pages and have encountered online threats or security problems. Their experiences provide useful and meaningful insights for understanding the types of cyber threats, the causes of digital vulnerability, and the challenges faced by small entrepreneurs in protecting their online assets.

2.4 Research Data and Analysis Plan

The study adopts a qualitative approach to understand experiences, threats, challenges, and organizational strategies for securing their digital property from cyber threats. It will focus on identifying common trends and themes in respondents' answers, offering a broad perspective on the issues faced by digital defenders at the forefront of the fight against cybercrime.

Data Collection and Preparation

Information will be gathered through open-ended questionnaires and semi-structured interviews with carefully selected participants, including information technology staff, cybersecurity officers, and organizational representatives who are directly involved in protecting digital assets. All interviews will be recorded with consent, transcribed verbatim, and organized systematically to guarantee accuracy and validity of the information. Before analysis, the researcher will undertake an initial process of data familiarization, reading and re-reading the transcripts to fully understand the participants' views and experiences.

Data Analysis and Procedure

The research will use thematic analysis, as described by Braun and Clarke (2006), to systematically identify, examine, and make sense of patterns within the qualitative data. Ensuring Credibility and Trustworthiness

Integration and Interpretation

The analyzed data will clarify the study's goals and be supported by existing literature on cybersecurity and digital defense. Combining findings will provide meaningful insight into threats, challenges, and the countermeasures organizations use to secure their digital assets.

2.5 Research Ethical Standards

The researchers maintained complete intellectual property rights protection throughout the research process. All data, information, and resources used in the study, such as articles, reports, and electronic data, were duly acknowledged and cited to credit their original authors. Plagiarism, unauthorized use, or replication of any patented or copyrighted material is not included in the research. Any data analysis tools, frameworks, or software used were open source, licensed correctly, or used with permission from the respective owners.

A. Protection of Intellectual Property Rights (IPR)

The researchers maintained complete intellectual property rights protection throughout the research process. All data, information, and resources used in the study, such as articles, reports, and electronic data, were duly acknowledged and cited to credit their original authors. Plagiarism, unauthorized use, or replication of any patented or copyrighted material is not included in the research. Any data analysis tools, frameworks, or software used were open source, licensed correctly, or used with permission from the respective owners.

B. Informed Consent

For this research, "Digital Defenders at the Frontline: Evaluating the Problems and Threats Experienced by Organizations in their Digital and Cyber Assets," informed consent was an ethical principle. Participants were adequately informed about the purpose, aims, procedures, and importance of the research before participating. They were aware of the potential risks and benefits, and written consent was obtained to ensure they entered the study voluntarily. Members were also promised that they would enter the research voluntarily and could leave at any time without penalty.

C. Data Privacy and Confidentiality

Researchers remained steadfast in adhering to data protection law and institutional policy to maintain participants' confidentiality. Personal and organizational information gathered from surveys and interviews was kept strictly confidential. Data were anonymized to avoid tracing individual responses back to identifiable participants. Collected data were stored securely in password-protected electronic files, accessible to the researchers alone, and for academic purposes alone.

D. Voluntary Participation and Freedom to Withdraw

The present research was completely voluntary. The volunteers were free to choose whether to participate in the study. They were also made aware of their right to withdraw from study at any time without suffering any ill consequences. In this way, all the volunteers participated voluntarily and with complete information concerning their rights.

E. Minimization of Harm and Risk Management

The researchers made careful efforts to minimize any potential harm or discomfort to the participants. Because the subject matter is cybersecurity attacks that could inadvertently cross limits of sensitive organizational issues, participants were informed that no personal or institutionally identifiable data would be revealed. Questions that could be stressful, embarrassing, or reputation-indamnifying were avoided in the study. Responding on a comfort level was encouraged, and all interactions were professional and respectful.

F. Beneficence and Contribution to Knowledge

This study was crafted to make significant contributions to the academic world and society. The results hope to open eyes to cybersecurity issues and help organizations enhance their cyber defense strategies. By identifying actual problems and threats, the study provides insights that can strengthen policies, improve digital security, and make technology more responsible.

G. Justice and Fair Participant Selection

Participants were selected fairly based on their relevance to the study's aims, particularly individuals employed in information technology, cybersecurity, or organizational management roles in the digital age. The process of participant selection was inclusive and aimed to reduce any gender, role, or institutional bias or discrimination. This approach upheld the ethical trusteeship of justice because all qualified individuals had an equal opportunity to participate.

H. Data Integrity and Accuracy

To preserve the credibility and reliability of the results, standardized measures were implemented for data collection and analysis. The authors of the report consistently made every effort to ensure that data from the observations were obtained and recorded accurately and reliably. Recognizing that there are potential limitations, biases, or errors in interpretation, one must openly acknowledge and declare these limits to maintain academic honesty and a valid conclusion.

I. Transparency and Honesty in Reporting

The researchers pledged to report all findings honestly, regardless of whether they supported the initial assumptions. No matter the form of information, literature, or materials included in the study, all data were correctly referenced and credited to avoid plagiarism. This commitment to honesty enhances the study's integrity and allows other researchers or practitioners to validate the findings.

J. Use of Patented or Copyrighted Materials

All copyrighted and patent materials utilized in this research, including published articles, pieces of software, and digital tools, were either fully cited or used with permission where necessary. The researchers did not use or reproduce others' intellectual property without permission. Regarding the open-source materials used, the researchers acknowledge that they continue to be used actively in research, while adhering to academic and legal conventions.

K. Considerations for Human Subjects

Throughout the research process, human participants were treated with respect, fairness, and dignity. Ethical protections were implemented to specifically protect participants by ensuring their right to privacy, their welfare, and their rights as participants in research approved by the Saint Michael College of Caraga. Anonymity was protected by not collecting any personal identifiers during data collection.

L. Ethical Clearance and Institutional Approval

Ethical Clearance and Institutional Approval before the conduct of the research study, ethics clearance and approval were obtained from the Ethics Review Committee of Saint Michael College of Caraga. The study adhered to the institution's ethical principles, the national data privacy statute, and international research ethics guidelines. This one ensured that the entire research enterprise proceeded reliably, transparently, and with professional and academic integrity.

M. Responsible Use of AI and Other Technologies

If any AI tools or digital platforms were used in the data analysis, transcription, or organization process, they were used responsibly to maintain the integrity and authenticity of the data. In every case, researchers verified AI-assisted results through human judgment and did not engage in data manipulation, bias, or any ethical breaches.